

Total No. of Questions : 6]

SEAT No. :

P112

[Total No. of Pages : 2

Oct.-16/B.E./Insem.- 170
B.E. (Information Technology)
NATURAL LANGUAGE PROCESSING (Elective - II(e))
(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Solve any 1 out of Q1 or Q2, Q3 or Q4 and Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) Draw and explain different levels of language understanding in natural language processing. Give example for each level. **[10]**

OR

Q2) a) Draw and explain flow of information in natural language understanding system. **[6]**

b) Consider the following context-free grammar:

S → NP VP N → dog V → sees

NP → Det N N → cat V → hates

VP → V N → mouse V → sneezes

VP → V NP Det → the

Which of the following sentences are recognised by this grammar, and why? **[4]**

- i) the dog sneezes the cat
- ii) the mouse hates
- iii) the cat the mouse hates
- iv) the mouse hates the mouse

P.T.O.

Q3) a) Differentiate between the depth-first and breadth-first to down parsing. **[5]**

b) Consider the following CFG:

$S \rightarrow NP V$

$S \rightarrow NP AUX V$

$NP \rightarrow ART N$

Trace one of the chart parsers in processing the sentence

1 The 2 man 3 is 4 laughing 5

with the lexicon entries: the: ART

man: N

is: AUX

laughing: V

Show every step of the parse, giving the parse stack, and drawing the chart each time a nonterminal constituent is added to the chart. **[5]**

OR

Q4) a) Explain reformulating grammar rules using logic programming with an example. **[5]**

b) Explain Top-Down chart parsing algorithm with an example. **[5]**

Q5) Draw and explain parsing with features in chart parsing algorithm. **[10]**

OR

Q6) a) What is augmented grammar? How is it useful in natural language processing? **[5]**

b) Explain some basic feature systems for English with an example. **[5]**

